Answers

Worksheet

Expressing multiplicative relationships as ratios and fractions

Task A: Multiplicative relationships as ratios

1) Using the ratio table, fill in the blanks.

Flour (g)	Eggs	Water (ml)	Sugar (g)
100	1	25	200

a) Number of eggs × 100 = Flour (g)

For every egg, there is 100 g of flour

b) Amount of water (ml) × 8 = Sugar (g)

For every ml of water, there is 8 g of flour

c) Amount of sugar (g) × 0.5 = Flour (g)

For every 1 g of sugar, there is 0.5 g of flour

2) Using the following bar models, fill in the gaps.

S	S	S	S	S	S	T

a) Shorts × ...3....... = T-shirts

For every pair of shorts, there are ....3 T-shirts....

C	C	M	M	M	M

b) Cat × ...2....... = Mice

For every cat, there are ....2 mice....

H	H	P	P	P	P

c) House × ...2....... = Phone

For every house , there are ....2 phones

Task A: Multiplicative relationships as ratios

3) Jacob spills ink all over his work.

Can you work out what was under the ink splats?

Here is a bar model showing the ratio water, glue and paper.

W	W	G	G	G	G	G	G	P	P	P

Ratio table

W	G	P
2	6	3

For every 1 part water, there are 3 parts glue.

For every 1 part paper, there are 2 parts glue.

For every 1 part glue, there are 0.5 parts *paper*.

For every *1 part water* there are 1.5 parts paper.

4) Laura, Lucas Alex and are making papier mache.

The instructions are: for every 4 tablespoons of white flour use 240 ml of water.

Tablespoons of flour	Water (ml)
4	240

$240 \div 4 = 60$

Tablespoons of flour × 60 = ml of water

Laura uses 8 tablespoons of flour and 480 ml of water.

Tablespoons of flour	Water (ml)
8	480

$480 \div 8 = 60$

Laura has the correct ratio. For every tablespoon of flour, there are 60 ml of water

Alex used 6 tablespoons of flour and 242 ml of water.

Tablespoons of flour	Water (ml)
6	242

$242 \div 6 = 40\frac{1}{3}$. *Alex has added 2 tablespoons and then added 2 ml. For every tablespoon of flour, there are* $40\frac{1}{3}$ *ml of water*

Lucas used 2 tablespoons of flour and 238 ml of water.

Tablespoons of flour	Water (ml)
2	238

$238 \div 2 = 119$. *Lucas has subtracted 2 tablespoons and then subtracted 2 ml. For every tablespoon of flour, there are 119 ml of water.*

Lucas's ratio will be too watery.

Task B: Multiplicative relationships as fractions

1) Using the bar model, fill in the gaps.

a)

C	D	D	D

Fraction of dogs = $\frac{3}{4}$

Cat × *3* = Dog For every cat, there are 3 dogs

b)

P	P	H	H	H	H	H

Fraction of hearts = $\frac{5}{7}$

Present × $\frac{5}{2}$ = Heart For every present, there are $\frac{5}{2}$ hearts

c)

A	O	A	O	O	O

Fraction of oranges = $\frac{4}{6}$

Apple × *2* = Orange For every apple, there are 2 oranges

2) There are four types of smoothies for sale all containing strawberries, apples and banana.

A For every strawberry, there are 2 slices of apples and 5 slices of banana.

B All fruit is equally split.

C Banana × $\frac{1}{3}$ = Apples. For every strawberry, there are 2 slices of banana.

D Apple and banana have the same ratio. Strawberry is 50% of banana.

A:

S	A	B
1	2	5

B:

S	A	B
1	1	1

C:

S	A	B
3	2	6

D:

S	A	B
1	2	2

a) Jacob likes strawberries, which smoothie should he pick? *B*

Smoothie B has the highest proportion of strawberries. $\frac{1}{3}$ are strawberries.

b) Sofia likes apple, which smoothie should she pick? *D*

Smoothie D has the highest proportion of apples. $\frac{2}{5}$ are apples.

c) Which smoothie has the least banana? *B*

Smoothie B has the least proportion of bananas. $\frac{1}{3}$ are bananas.

d) Create your own smoothie so the ratio of banana > ratio of apple > ratio of strawberries.